



INTU: The Invisible Co-Pilot

Transforming Cabin Intelligence from Active Input to Passive Reception

We are redefining the Human-Machine Interface by removing the interface entirely.

The "Smart" Car is Too Needy

The Cognitive Tax of Modern Driving



Active Input Overload

Modern cars demand **Active Input**: Touchscreens, voice commands, and complex menus distract the driver.

Mental Fatigue

Cognitive Load: For long-haul truck drivers, every interaction adds mental fatigue.

Waiting for Commands

The Gap: Cars have sensors, but they lack intuition. They wait for commands instead of anticipating needs.

From Operator to Passive Receptor

The Paradigm Shift

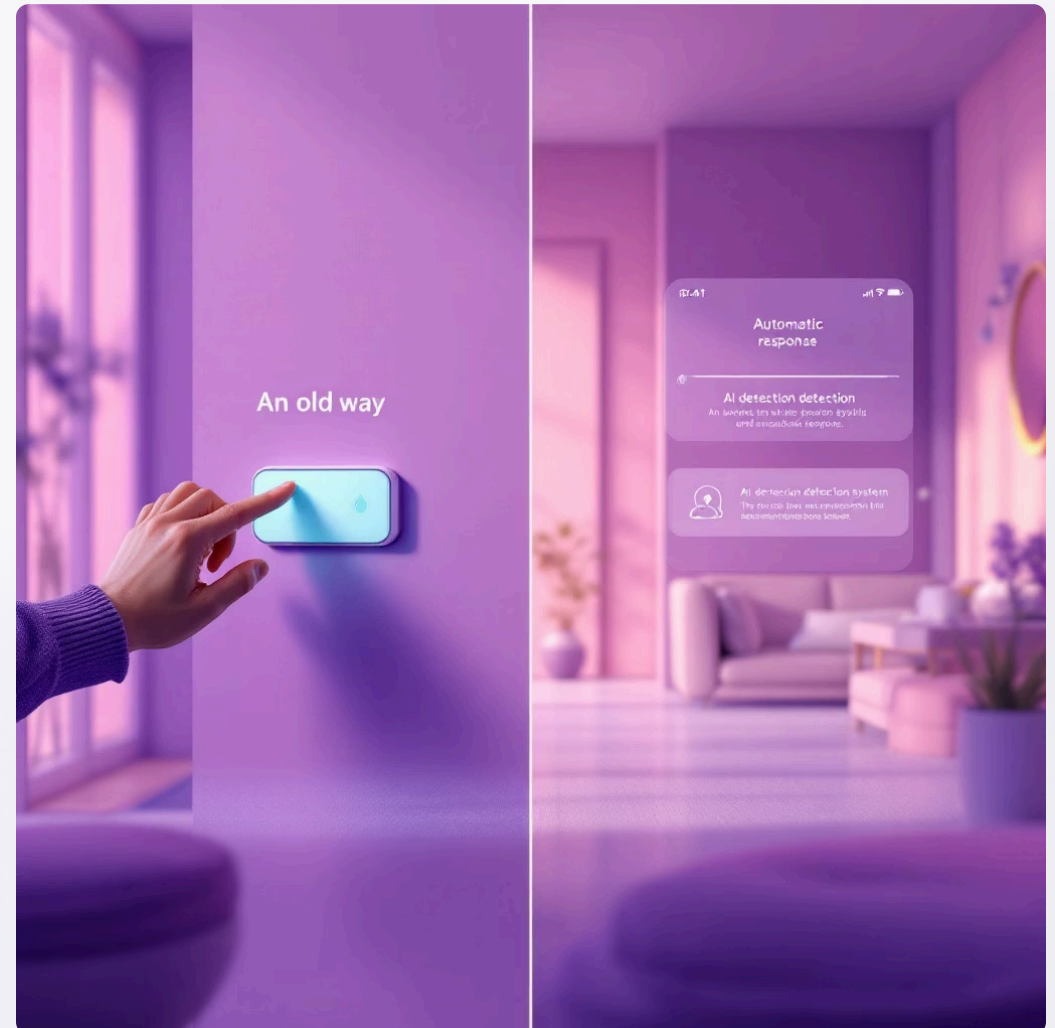
INTU flips the script. The driver doesn't talk to the car; the car understands the driver.

Zero-UI

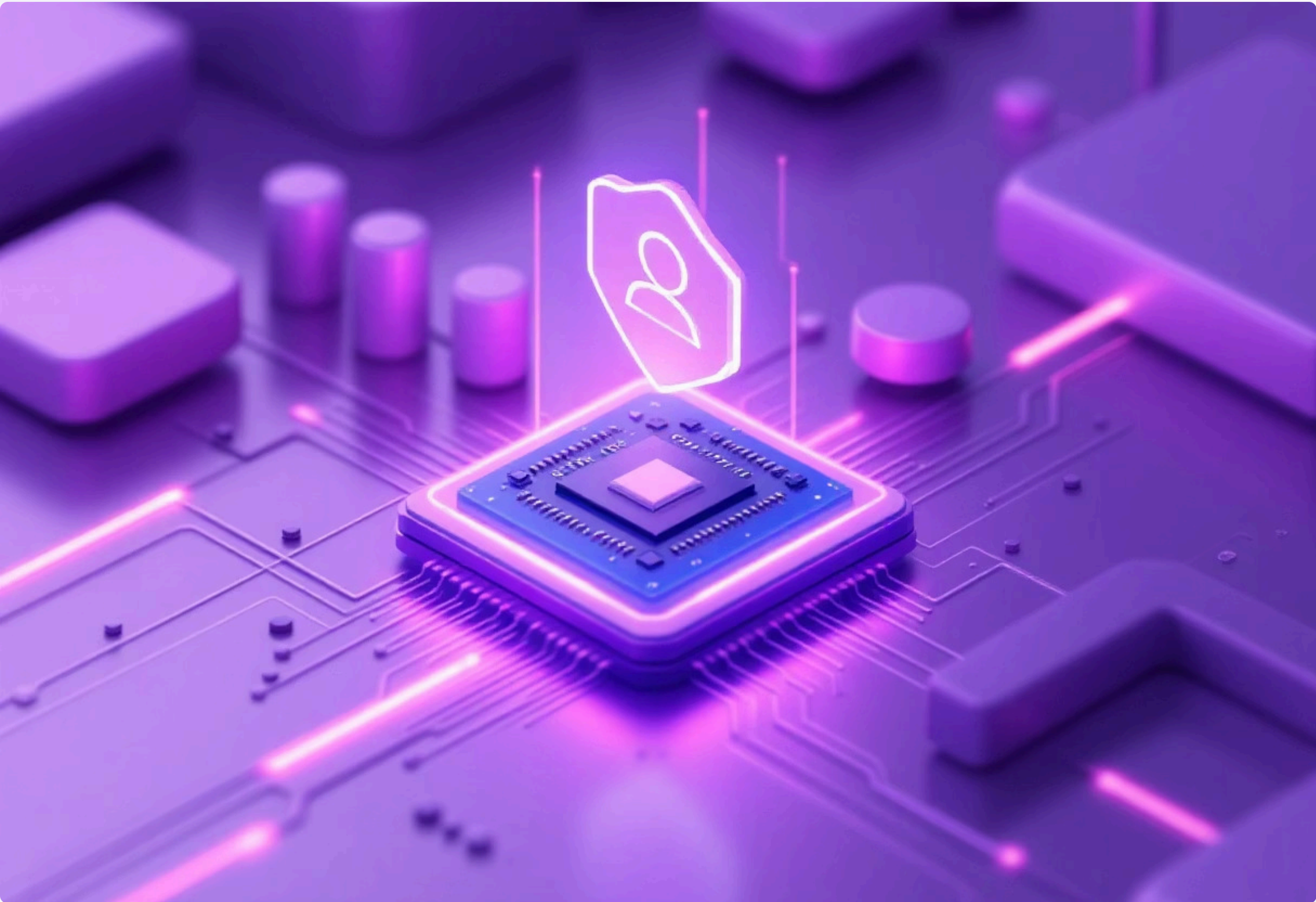
No buttons, no wake words. Just seamless, invisible support.

Core Promise

Detecting subconscious needs—posture, gaze, and stress—and reacting *before* the driver realizes they need help.



Powered by Edge AI. Built on STM32.



Hardware

Runs entirely on a standard **STM32 MCU** with a **Single RGB Camera**.



Privacy First

100% On-Device processing. No video leaves the cabin. No cloud dependency.



Performance

Real-time inference (<50ms latency) optimized for low power consumption.

 **The Feasibility:** Enterprise-grade AI running on automotive-standard hardware.

Posture Detection Engine

Real-time driver state analysis using YOLO v8/v11

Performance Metrics

60%

CPU Load

Abundant safety margin on system resources.

39ms

Inference Time

Near real-time processing for instant feedback.

25

FPS

Consistent frame rate for smooth monitoring.

0.98MB

RAM Usage

Efficient memory footprint on embedded systems.

3.17MB

Flash Usage

Efficient storage on embedded systems.

99%

cos similarity in public dataset

High accuracy across diverse conditions.

Not Just Detection. Understanding.



We go beyond simple object detection. We built a **Contextual Rules Engine**.

Fusion Strategy

Combining Gaze + Head Pose + Action Recognition to understand *intent*.

Example: A slouch at a red light is "Relaxation." A slouch at 60mph is "Fatigue." INTU knows the difference.

Unlocking the Value of Premium Hardware

Business Value: The Hardware Multiplier



Forvia's Opportunity

INTU turns passive hardware (Seats, Lights) into active health devices.



The Upsell

It creates a compelling reason for fleet owners to buy premium "Wellness Packages"—because the software ensures they are actually used.



Revenue Model

A low-cost software layer that drives high-margin hardware sales.

Premium Hardware

+ INTU AI =

High User ROI

Feasible. Scalable. Ready.

Scalability & Market Adoption

Cost Effective

No expensive GPUs needed. Uses existing DMS cameras mandated by regulations (Euro NCAP).

Market Scale

Modular design allows deployment from Economy cars (Safety features only) to Luxury Fleets (Full Wellness suite).

Roadmap

Phase 1

Trucking Logistics (High Pain Point)



Phase 2

Mass Market Passenger Vehicles



CONCLUSION

The Future is Intuitive

We are building the bridge between human needs and machine capabilities.

INTU: Protection you can feel, not just see.



Made with GAMMA

INTU: Action Before Interaction

The subconscious co-pilot for professional drivers.

Call to Action

Let's make the "Cockpit of the Future" a reality today.

